SMART INDIA HACKATHON 2026

- **Problem Statement ID –** 25132
- **Problem Statement Title -** Smart Ambulance Corridor:

Intelligent Traffic Management for Emergency Response

- **Theme -** Miscellaneous
- **Category-** Software
- **Team Name - resQ Siren**

THE PROBLEM STATEMENT

Ambulances lose 15–20 critical minutes navigating Indian traffic — minutes that decide survival in cardiac arrest and trauma cases.

Current traffic infrastructure runs on fixed signal timing with zero visibility into approaching emergency vehicles. Existing "green corridor" efforts rely on manual police coordination or basic vehicle detection — and break down when it matters most.

Where it breaks down

No severity awareness
Every ambulance is treated the same, whether it's carrying a cardiac arrest patient or a stable transfer.

No conflict resolution
Systems assume one ambulance at a time; real cities have multiple emergencies converging on the same junction.

No verification
Signal-priority requests can't be authenticated, leaving them open to spoofing and misuse.



Choking life: 12 on way to hospitals die in traffic jams

TNN / Updated: Oct 18, 2016, 09:49 IST

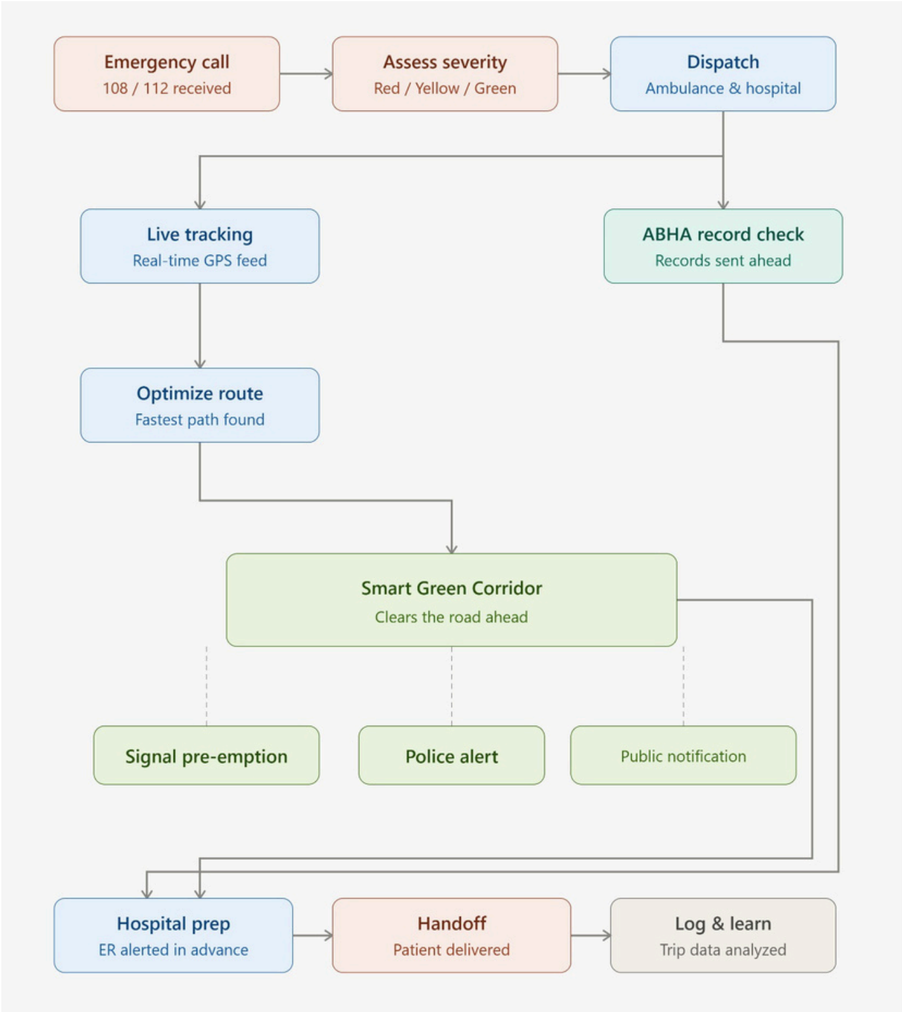
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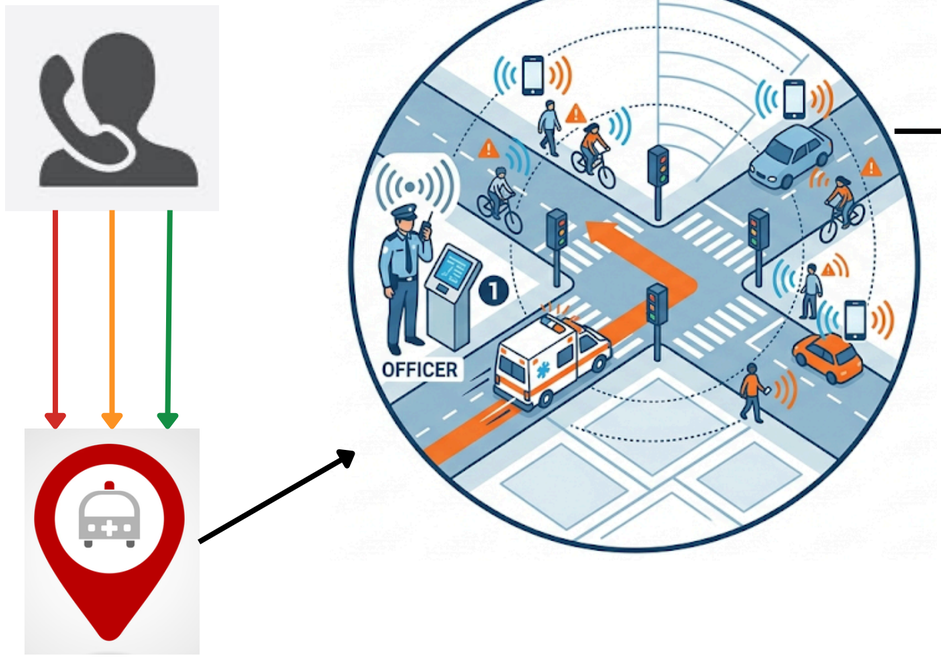
At least a dozen people, especially those injured in road accidents, lost their lives in last six months in the city, while on the way to hospitals.

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Smart Ambulance Corridor: Intelligent Traffic Management for Emergency Response

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A dispatcher-triggered, severity-weighted, authenticated traffic corridor that dynamically routes ambulances and arbitrates between competing emergencies at shared junctions. Severity is generated once at call intake and drives every downstream decision — routing, signal alerts, priority, and post-trip performance tracking.

TECHNICAL APPROACH

Frontend
- React.js + Tailwind CSS
- React Native / Flutter (driver app)

Backend
- Node.js (NestJS) / FastAPI
- REST APIs, dispatch + severity logic

Real-Time Layer
- Socket.IO / WebSockets
- JWT + HMAC dispatch signing

Database
- PostgreSQL (trips, records)
- Redis (live geospatial data)

Maps & Routing
- Google Maps / Mapbox API
- OSRM (self-hosted fallback)

Signal Integration
- Mock ITMS junction service
- RFID / GPS-geofence (future)

Notifications
- Firebase Cloud Messaging
- Twilio SMS gateway

Hosting & Infra
- Vercel / Render / Supabase
- GitHub Actions CI/CD

FEASIBILITY AND VIABILITY

Feasibility of the Idea

- Technically feasible — built on proven tech, mock signal layer designed to plug into real hardware later
- Operationally feasible — no new field hardware needed; rolls out in phases
- Cost feasible — open-source and low-cost cloud tiers, costs scale with usage

Potential Challenges & Risks

- Traffic signal hardware varies widely across cities and vendors
- Health record access raises data privacy and consent concerns
- Rollout needs coordination across the 108 control room, police, and hospitals

Strategies to Overcome Them

- Mock ITMS layer now, with a pluggable adapter design for real hardware
- Consent-based ABHA tokens only — no raw patient health data stored
- One shared, role-based dashboard so all three agencies see the same data

IMPACT AND BENEFITS

How the Process Changes

TRADITIONAL 108 RESPONSE	SMART AMBULANCE CORRIDOR
Coordination relies on phone/radio calls between control room, ambulance, and police	One dashboard shares live, verified status with all three parties at once
Traffic police react to an ambulance as they notice it approaching	Automated advance alert reaches the junction before the ambulance does
Hospital learns patient details only when the ambulance arrives	Hospital gets an advance alert with severity code and medical history
No standard way to confirm a signal request is from a genuine ambulance	Every request is checked against a signed, authenticated dispatch record

Benefits of the Solution

Social

- Earlier hospital readiness for critical patients
- A more transparent, accountable 108 system

Economic

- Fewer wasted trips through better dispatch matching
- Usable trip-history data for resource planning

Environmental

- Traffic-aware routing avoids unnecessary detours
- Fewer emergency-driven traffic disruptions